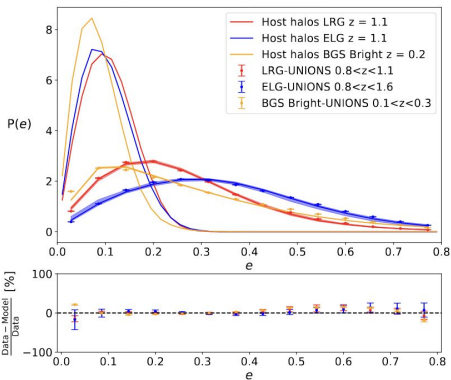


Inferring the 3D shapes of galaxies inside halos

with  **UNIONS**

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1 **Context:** galaxy-halo connection in 3D for intrinsic alignments

2 **Methodology:** simulation-based inference using a forward model of 3D shapes of galaxies in halos

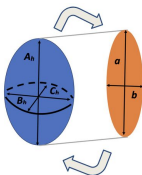
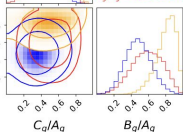
3 **Results:** constraints on 3D shapes for 3 galaxy samples from the distributions of ellipticities of galaxies and halos

$$\begin{aligned} C_g/A_g &= 0.48 \pm 0.22 \\ C_g/A_g &= 0.36 \pm 0.15 \\ C_g/A_g &= 0.50 \pm 0.16 \end{aligned}$$



LRG-UNIONS $0.8 < z < 1.1$
ELG-UNIONS $0.8 < z < 1.6$
BGS Bright-UNIONS $0.1 < z < 0.3$

$$\begin{aligned} B_g/A_g &= 0.82 \pm 0.11 \\ B_g/A_g &= 0.59 \pm 0.19 \end{aligned}$$



Takeaway: we can constrain the distribution of 3D shapes of galaxies from observations, using simulations.